

Aryan Miriyala

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PROFESSIONAL SUMMARY

Software engineer with experience building reliable backend/data workflows, secure internal tools, and AI-assisted engineering systems using **Python, SQL, MySQL, PostgreSQL, Oracle, AWS**, and **Docker**. Background includes production data automation, database validation, access control, reproducible developer environments, and technical tradeoff communication across operational systems.

EDUCATION

Bowling Green State University

Master of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Aug. 2024 – Aug. 2026

Bowling Green State University

Bachelor of Science in Computer Science, GPA: 4.00/4.00

Bowling Green, OH

Jan. 2021 – Apr. 2024

EXPERIENCE

Artificial Intelligence & Computer Science Intern

SmartSolve Industries

May 2026 – Present

Bowling Green, OH

- Built secure internal software with **Next.js, TypeScript, PostgreSQL, Drizzle ORM, SSO**, and auth middleware for onboarding and org-chart workflows around sensitive employee data.
- Created a cross-platform **Docker** devcontainer for **WSL2/Colima**, standardizing reproducible environments for AI-assisted development, debugging, implementation planning, and code review.
- Applied **Codex** and **Claude Code** to accelerate system design and feature iteration while validating generated output through human review and proprietary-code isolation.

Data Engineering Intern

American Association of Insurance Services

May 2025 – Aug. 2025

Lisle, IL

- Automated a manual **MySQL/SQL** billing workflow with **Python, PySpark**, and **AWS Glue**, processing **20+ TB** of production insurance data for **700+** member companies.
- Profiled **160+ MySQL/Oracle/Impala** tables with **Python, Pandas, SQL**, and JDBC to map source-system similarities and define **25** MDM domains for cloud ETL migration.
- Built ETL validation, reverse-ETL, and custom **AWS IAM** access patterns across **MySQL**-backed, **S3**, and MDM workflows, improving issue diagnosis and controlled data access.

Software Engineering Intern

American Association of Insurance Services

May 2023 – May 2024

Lisle, IL

- Engineered **Python/JSON** backend automation generating **1,000+** production **SQL** tables across **MySQL, Oracle**, and **PostgreSQL**, supporting standardized data models across **10+** insurance lines.
- Built **Python/Pandas** workflows with multiprocessing, fuzzy matching, and hashing to clean large CSV datasets for production data-standardization pipelines.
- Contributed to a **React/Node.js** modernization by implementing **RBAC** and **JWT** patterns, replacing broad employee-wide access with role-specific controls for roughly **110** internal users.
- Automated event-driven validation for **MySQL/Oracle/PostgreSQL** rows with **AWS Lambda**, tokenizing PII using **hashlib/boto3** and ingesting structured JSON into **S3**.

Software Engineering Intern

Alliance for Paired Kidney Donation

May 2022 – Aug. 2022

Toledo, OH

- Secured an **AWS-hosted Lucee/CFML** healthcare workflow platform with global **anti-CSRF** protection, IP-aware audit logging, and access-denied/404 routing across form-heavy internal workflows.
- Modernized **HTML5** date/time inputs, logistics defaults, Matchgrid import ordering, MFI upload paths, and duplicate-antigen crash handling to improve data-entry reliability for internal operations.

PROJECTS

HealthTrend Data Pipeline | *Kafka, Hadoop HDFS, Spark, PySpark, Docker*: Built a containerized healthcare data pipeline routing CSV patient records and JSON health events into batch and streaming processing paths.

Self-Adaptive Parallelism | *C++, OpenMP, Python, Docker, Linux*: Built a runtime research system using per-epoch telemetry and a UCB controller to tune scheduling policy, chunk size, and thread count.

RocketGrader | *Angular, TypeScript, Express, MongoDB, Auth0, AWS S3*: Built a full-stack AI feedback platform with REST APIs, role-based dashboards, file upload/storage, parsing, and LangChain/Mistral AI grading workflows.

Diff-Grounded PR Generation | *Python, OpenAI API, GitHub API*: Built an LLM pipeline generating evidence-grounded PR descriptions from commits, diffs, linked issues, and metadata with automated evaluation against code-change evidence.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript/JavaScript, C++, C#, C, Lucee/CFML

Backend, Databases & APIs: Node.js, Express, REST APIs, PostgreSQL, MySQL, Oracle, MongoDB, Drizzle ORM, JDBC

Cloud, Data & Systems: AWS Glue, Lambda, S3, IAM, PySpark, Pandas, Kafka, Hadoop HDFS, Docker, Git, Linux

Reliability, Security & AI: SSO, JWT, RBAC, Auth0, anti-CSRF, audit logging, PII tokenization, Codex, Claude Code,

OpenAI API, Mistral AI, LangChain